WEEK 7

CODE :

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <string.h>
int main()
{
    int fd;
    char message[100];
    fd = open("client_fifo", O_WRONLY);
    strcpy(message, "Hello Server");

    write(fd, message, strlen(message) + 1);
    close(fd);

    fd = open("server_fifo", O_RDONLY);
    read(fd, message, sizeof(message));
    printf("Server: %s\n", message);

    close(fd);
    return 0;

}
```

```
princy@Samuel :~$ nano week6_client.c
princy@Samuel :~$ gcc week6_client.c
princy@Samuel :~$ ./a.out
Server: Hello Server
```

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <string.h>
int main()
{
    int fd;
    char message[100];
mkfifo("client_fifo", 0666);
mkfifo("server_fifo", 0666);
printf("Server waiting...\n");
fd = open("client_fifo", O_RDONLY);
read(fd, message, sizeof(message));
printf("Client: %s\n", message);
close(fd);
fd = open("server_fifo", O_WRONLY);
strcpy(message, "Message processed by server");
write(fd, message, strlen(message) + 1);
close(fd);
return 0;
}
```

```
  GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
void handle_signal(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT received\n");
else if (sig == SIGTERM)
    printf("SIGTERM received\n");
else if (sig == SIGUSR1)
    printf("SIGUSR1 received\n");
}
int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);
    signal(SIGUSR1, handle_signal);
printf("Process ID: %d\n", getpid());
printf("Waiting for signals...\n");
while (1)
{
    sleep(1);
}
return 0;
}
```

```
princy@Samuel:~$ nano week6_server.c
princy@Samuel:~$ gcc week6_server.c
princy@Samuel:~$ ./a.out
Server waiting...
```

```
  GNU nano 8.7.1
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
void handle_signal(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT received\n");
else if (sig == SIGTERM)
    printf("SIGTERM received\n");
else if (sig == SIGUSR1)
    printf("SIGUSR1 received\n");
}
int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);
    signal(SIGUSR1, handle_signal);
printf("Process ID: %d\n", getpid());
printf("Waiting for signals...\n");
while (1)
{
    sleep(1);
}
return 0;
}
```

```
princy@Samuel:~$ nano week6_signal.c
princy@Samuel:~$ gcc week6_signal.c
princy@Samuel:~$ ./a.out
Process ID: 1720
Waiting for signals...
^C
SIGINT received
^C
SIGINT received
```